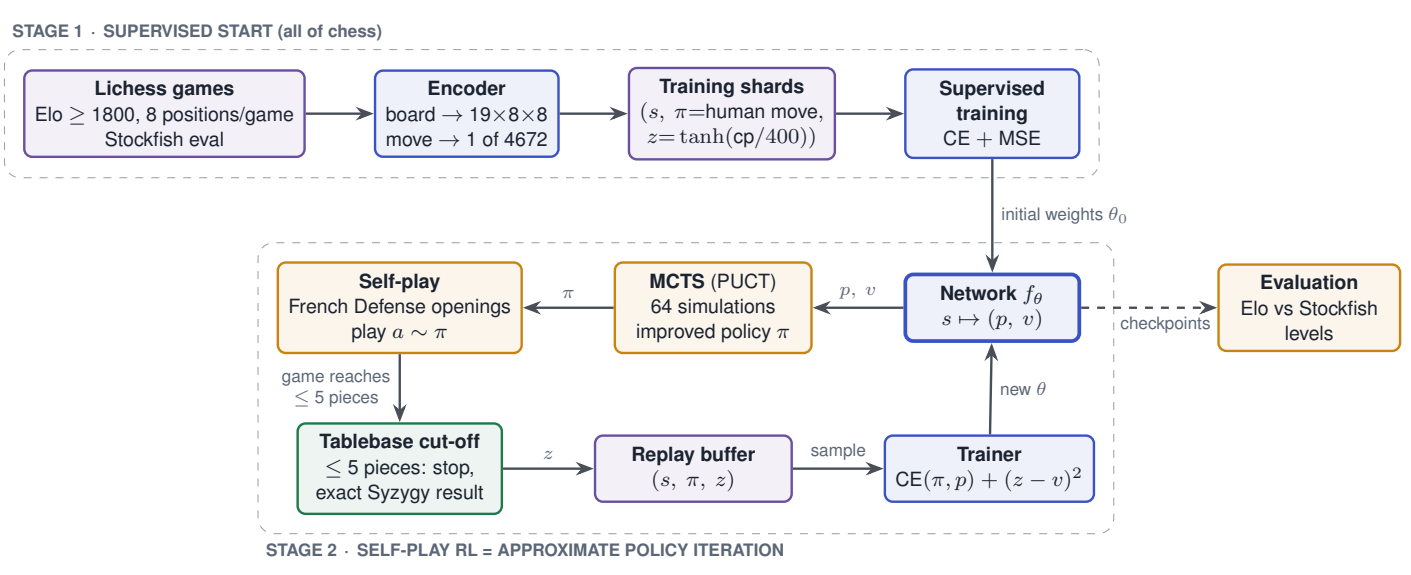

STAGE 1 · SUPERVISED START (all of chess)
Lichess games
Elo ≥ 1800, 8 positions/game
Stockfish eval
Encoder
board → 19×8×8
move → 1 of 4672
Training shards
(s, π=human move,
z= tanh(cp/400))
Supervised training
CE + MSE
initial weights θ0
Self-play
French Defense openings
play a ∼ π
π
MCTS (PUCT)
64 simulations
improved policy π
p, v
Network fθ
s ↦ (p, v)
checkpoints
Evaluation
Elo vs Stockfish levels
game reaches ≤ 5 pieces
new θ
Tablebase cut-off
≤ 5 pieces: stop,
exact Syzygy result
z
Replay buffer
(s, π, z)
sample
Trainer
CE(π, p) + (z − v)²
STAGE 2 · SELF-PLAY RL = APPROXIMATE POLICY ITERATION